

WORKSHOP BENCH PROJECT SUITE



THE BENCH — 35 SMALL WEB TOOLS

Complete Project Overview & Comprehensive Architectural Specification



PROJECT SUMMARY & STATS

- **Total Inventory:** 35 Self-Contained Web Applications
- **Infrastructure Promise:** 0 Servers / 0 Build Steps / 100% Client-Side LocalStorage
- **Technology Stack:** HTML5, CSS3, ES6+ JavaScript, GSAP Animations, Bootstrap 5, Parallax Scroller
- **Attribution:** Built by Vishu — "A bench of 35 small tools, made to be used, not just looked at."

1. 🎯 What This Is & Design Philosophy

"The Bench" (project directory: 20-web-projects) is a static, no-build, no-backend collection of 35 self-contained web applications hung on a single hub page styled like a workshop pegboard. Every tool is plain HTML, CSS, and JavaScript with no frameworks or bundlers. Data persistence relies entirely on the browser's localStorage under namespaced keys.

Core Engineering & Design Principles

- ⚡ **Zero Infrastructure:** No server, no build step, and no package manager required. Opening index.html directly in any browser unlocks 100% functionality.
- 🎨 **One Consistent Visual Shell:** Every tool page shares the same header navigation (<- The Bench), tool title block, button chips, glass-panel visual styling, and dark/light theme toggle.
- 🛠️ **Real Working Logic:** Each app implements genuine domain logic (e.g., scientific calculator evaluation, state-wise CGST/SGST/IGST tax splits, and Breadth-First Search dungeon solvability verification).
- 🔒 **Namespaced Local Storage:** Local persistence uses a shared helper (Bench.Storage) writing keys as bench:<namespace>:<name> to prevent key collisions across tools.
- 🌐 **Graceful Network Degradation:** All external API calls wrap in try/catch blocks with fallback toast alerts to guarantee offline usability.

2. 📁 Project Structure & Code Architecture

The entire suite maintains a strict, standardized file organization with zero extraneous asset folders per tool:

```
20-web-projects/
├── index.html           <- Hub Dashboard ("The Bench")
├── assets/
│   ├── css/
│   │   ├── style.css   <- Shared Design System (Colors, Type, Themes)
│   │   ├── loader.css  <- CSS Loading Animations & Skeleton Shimmers
│   │   └── parallax.css <- Parallax Hero Background Scroller
│   └── js/
│       ├── storage.js   <- Bench.Storage, Bench.toast, Bench.debounce
│       └── main.js      <- Catalog Data, Search, Filter & GSAP Animations
└── projects/
    ├── 01-todo-list/ ... 35-expense-invoice-tracker/
    └── [Each project contains exactly: index.html, script.js, style.css]
```

3. The Full Catalog of 35 Web Tools

Below is the complete inventory of all 35 tools registered in assets/js/main.js categorized across 9 distinct functional domains:

#	Tool Title	Category	Key Features & One-Line Pitch
01	 Task Board	Productivity	Categorized tasks, priority tags, drag-to-reorder, saved filters
02	 Calculator	Utility	Standard and scientific modes with a running history drawer
03	 Snake Game	Game	Canvas arcade snake with skins, sound toggle, local high-score table
04	 Sign-Up Flow	Form	Multi-step registration with live validation and password-strength meter
05	 Hero Slider	Media	Touch-swipe slider with autoplay, captions, thumbnail rail
06	 BMI Gauge	Utility	Interactive sliders feeding a live gauge and saved history log
07	 Password Smith	Utility	Entropy-rated password generator with a small saved vault
08	 Palette Forge	Utility	Lock colors, convert HEX/RGB/HSL, export a saved palette
09	 Counter Bank	Productivity	Multiple named counters with step size, targets, and history
10	 QR Bench	Utility	Custom QR codes with color/size control and a scan history log
11	 Weather Bench	Data	City search with a 5-day outlook and saved favorite locations

#	Tool Title	Category	Key Features & One-Line Pitch
12	 Ledger	Finance	Income vs. expense bars, category filters, monthly summaries
13	 Quiz Master	Game	Timed trivia with card-flip reveals and a top-scores board
14	 Focus Timer	Productivity	Configurable work/break cycles with streak tracking
15	 Memory Flip	Game	3D flipping match game across three grid sizes with move counter
16	 Markdown Bench	Productivity	Live-preview note editor with tags, word count, and export
17	 Typing Trial	Game	WPM and accuracy trainer with a live results graph
18	 Recipe Box	Data	Ingredient search, cook timers, bookmarked favorites
19	 Unit Bench	Utility	Currency, length, weight, temperature, and speed conversions
20	 Sound Deck	Media	Playlist player with an animated visualizer and equalizer
21	 Car Racing	Game	Lane-dodging racer that speeds up the longer you survive
22	 Treasure Hunter	Game	Procedural dungeon crawl — BFS verified solvable layout
23	 Space Defender	Game	Wave-based invader shooter with escalating enemy fire
24	 Zombie Arena	Game	Twin-stick survival against endless waves of the horde
25	 Tower Defense	Game	Place towers, hold the path, survive escalating waves
26	 Gradient Generator	Creative	Color-theory palettes turned into ready CSS gradients
27	 Logo Maker	Creative	Emblem, name, and layout combined into an exportable logo
28	 Poster Designer	Creative	Drag-to-place headline and subtitle over custom background
29	 Pixel Art Studio	Creative	Grid-based pixel painter with fill tool and PNG export
30	 Social Creator	Creative	Square, story, or landscape posts, ready to export
31	 Receipt Maker	Business	Itemized receipts with tax, ready to print or save as PDF
32	 Invoice Generator	Business	Client invoices with line items, tax, and due dates
33	 GST Billing	Business	Indian GST invoices with automatic CGST/SGST/IGST split
34	 Quotation Maker	Business	Client estimates with validity dates and clear terms

#	Tool Title	Category	Key Features & One-Line Pitch
35	 Expense Tracker	Business	Invoiced, collected, pending, and expenses in one dashboard

4. External API & Network Dependencies

Of 35 tools, 29 are completely network-independent. The remaining 6 handle external network calls gracefully:

-  **Weather Bench (#11):** Open-Meteo geocoding & forecast API (no key needed). Shows toast alert and defaults to saved favorites or Rajkot offline.
-  **Recipe Box (#18):** TheMealDB free API. Favorited recipes maintain local JSON snapshots for offline viewing.
-  **Unit Bench (#19):** open.er-api.com USD exchange rates. Silently falls back to a 1:1 USD rate table if offline.
-  **QR Bench (#10):** Loads qrcode.js library via CDN. Requires network on initial load.
-  **Hero Slider (#05):** Picsum Photos placeholder images. Autoplay and navigation function fine offline.
-  **Hub Shell:** Loads GSAP & Google Fonts via CDN. Gracefully falls back to standard CSS animations and system fonts.

5. Deep-Dive Implementation Details

-  **Safe Math Evaluation (#02):** Expressions evaluate cleanly via `Function("use strict"; return (...))()` after strict sanitization.
-  **Web Audio API Synthesis (#03, #20):** Sound effects generate in real-time via Web Audio API Oscillator Nodes (zero external MP3 assets needed).
-  **Dungeon Solvability Verification (#22):** Treasure Hunter validates every procedurally generated map via Breadth-First Search (BFS) to guarantee a solvable path.
-  **Native Print-to-PDF (#31-#34):** Business tools utilize CSS `@media print` styling with `window.print()` for pixel-perfect PDF exports without heavy libraries.
-  **Indian GST Auto-Split (#33):** GST Billing compares seller/buyer state strings to dynamically split tax between CGST + SGST (intra-state) or IGST (inter-state).

6. 🚀 Developer Extension Guide (Adding Tool #36)

🔧 STEPS TO ADD A NEW TOOL

1. **Create Folder:** Create `projects/36-your-slug/` with `index.html`, `script.js`, and `style.css`.
2. **Set Storage Namespace:** Define `const NS = 'bench:tool36'` inside an IIFE in `script.js`.
3. **Register in Catalog:** Append your tool object to `Bench.catalog` in `assets/js/main.js`.
4. **Deploy:** Refresh `index.html` — your tool will immediately appear, filter, and search with zero build steps!

The Bench — 35 Small Web Tools Complete Project Overview & Technical Documentation • Generated for User Workspace